

CYBER SECURITY COURSE

Advanced Nmap Assessment of Windows Server

Professional Case Report 18

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Case ID	CSC-2026-18
Discipline	Network assessment
Overall risk	Critical
Evidence status	Laboratory methodology supported by saved Nessus evidence

Purpose

Explain Windows TCP/IP scan behavior and build a repeatable assessment that balances depth, speed and evidence quality.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

Metasploitable 3 Windows Server 2008 R2 was assessed with advanced Nmap discovery, flag manipulation, firewall-evasion and NSE techniques. The assessment showed that scan interpretation matters as much as command selection. Windows-specific responses and verified SMB exposure guided the final risk priority.

Case Profile

Scenario	Metasploitable 3 Windows Server 2008 R2 was assessed with advanced Nmap discovery, flag manipulation, firewall-evasion and NSE techniques.
Environment	Kali Linux against Metasploitable 3 Windows Server 2008 R2 in an isolated VirtualBox network.
Tools	Nmap, NSE, XML output, SMB enumeration, OS fingerprinting

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Used TCP, UDP, SCTP and ICMP probes for host discovery.
2. Compared FIN, Maimon and Window scans against the Windows stack.
3. Tested fragmentation, decoys, source-port selection and MAC spoofing.
4. Controlled rate, retries and parallelism.
5. Ran SMB and vulnerability scripts and exported XML results.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Windows flag behavior	Medium	Uniform RST responses limit the value of FIN and related stealth scan for open-port differentiation.
F02	SMB exposure	Critical	Saved Nessus evidence confirmed MS17-010 on the target, making S a priority validation path.
F03	Obsolete platform	Critical	Windows Server 2008 R2 was unsupported and exposed multiple leg services.
F04	Rate and output control	Medium	Explicit timing limits and XML output improve reproducibility and red accidental disruption.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
nmap -sn -PS445 -PU137 <target_ip>
nmap -sS --mtu 16 <target_ip>
```

```
nmap -sV --script vulners <target_ip>
nmap -O --osscan-guess <target_ip>
nmap -sV -p- -oX scan_results.xml <target_ip>
```

Risk Assessment

Risk	Level	Operational impact
Remote SMB compromise	Critical	Unauthenticated code execution and lateral movement
Evasion technique misuse	High	Fragmentation and decoys can reduce simple detection

Recommendations

1. Retire unsupported Windows systems.
2. Patch SMB and restrict port 445 between segments.
3. Tune IDS for fragmented and decoy scan patterns.
4. Store XML output with timestamps for comparison after remediation.

Knowledge Demonstrated

- Advanced Nmap
- Windows stack analysis
- Firewall evasion
- Structured scan output

Conclusion

The assessment showed that scan interpretation matters as much as command selection. Windows-specific responses and verified SMB exposure guided the final risk priority.

Hands On Lab Practical 18

Completed as part of the authorized cyber security course practical work.